

CSE427 | Machine Learning**Quiz: 1**

Name:	ID:	Sec:
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You are given data from an online gaming platform.

ID	Game Mode	Hours Per Week	Team Play	Premium Player
1	Co-op	2	No	No
2	Solo	5	No	No
3	Multiplayer	8	Yes	Yes
4	Solo	3	Yes	No
5	Multiplayer	7	Yes	Yes
6	Co-op	10	Yes	Yes
7	Solo	4	Yes	No
8	Co-op	9	Yes	Yes
9	Multiplayer	6	No	No
10	Solo	11	Yes	Yes

Find the Missing values, show complete working for each missing value. Find the root node for the classification tree using Information Gain.

Hours Per Week	Gini	Information Gain
2.5	0.444	0.108
3.5	0.375	(d)
(e)	0.286	0.396
5.5	(c)	0.610
(g)	(a)	(b)
7.5	0.167	0.610
(f)	0.286	0.396
9.5	0.375	0.237
10.5	0.444	0.108